

April 2013

CE210 MICROPROCESSOR & COMPUTER ARCHITECTURE

Date: 27.04.2013, Saturday Time: 01:30 p.m. To 04:30 p.m. Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1 Answer the following questions. [11]

- [A] Microcontroller is a "Computer on a Chip". Explain why? [1]
- [B] Assembly language is processor independent. State true/false and justify. [2]
- [C] Explain the support available in 8085 to interface slower devices. [2]
- [D] Which flag bit in 8085 cannot be tested using conditional branch instruction? [2]
- [E] If 8085 is operating at 1 MHz clock, how much time 8085 will take to transfer one byte information to peripheral device connected as I/O mapped I/O? [2]
- [F] How many machine cycles MOV M,A instruction will take? Name each one. [2]

Q-2 Answer the following questions. [12]

- [A] 2100 LXI SP, 3000H [5]
LXI B, 2300H
PUSH B
POP D

Draw the stack frame neatly after the execution of the POP D instruction. Specify the content of D and E register

- [B] 8085 system requires following memory map in the system : [7]
EPROM : 16 Kbytes at processor address space at 0000h
RAM : 16 Kbytes at processor address space 8000H
8 Kbytes RAM and EPROM devices are available. Design for absolute decoding (partial decoding not allowed). Draw the neat circuit diagram..

OR

- [B] What is DMA? Explain its interfacing with 8085 in detail. [7]

Q-3 Answer the following questions.(Attempt any Two) [12]

- [A] 2100 MVI A,55H [12]
Describe the content of MSB address bus (A8 to A15) and data bus (AD0 to AD7) for every T state during the execution of the above instruction.
- [B] Write a program to generate 1 KHz square wave on PC7 pin of 8255. 2 MHz crystal is connected to 8085. Draw neat flow chart and state your assumptions very clearly.
- [C] Draw and explain Functional Block Diagram of 8085 microprocessor.
- [D] Draw and explain timing diagram of PUSH instruction.

SECTION-II

Q-4 Answer the following questions. [11]

- [A] How does a microprocessor differentiate between data and instruction code? [2]
- [B] Explain Instruction: LXI H,2300H, MOV A,M, STA 2300H [3]
- [C] State the physical addressing capacity of 8086, 80286 and 80386 processor. [3]
- [D] Draw the circuit diagram to demultiplex LSB address from data bus from 8085. [3]

Q-5 Answer the following questions. [12]

[A] Write an assembly program to count number of '1's in a given 8-bit number. (eg 41H contains two '1's). [5]

[B] Explain the function of the following instructions with example : [7]

(1) CPI 55H (2) DAD H (3) LDAX D (4) RAL (5) RLC (6) RET (7) CALL 2300H

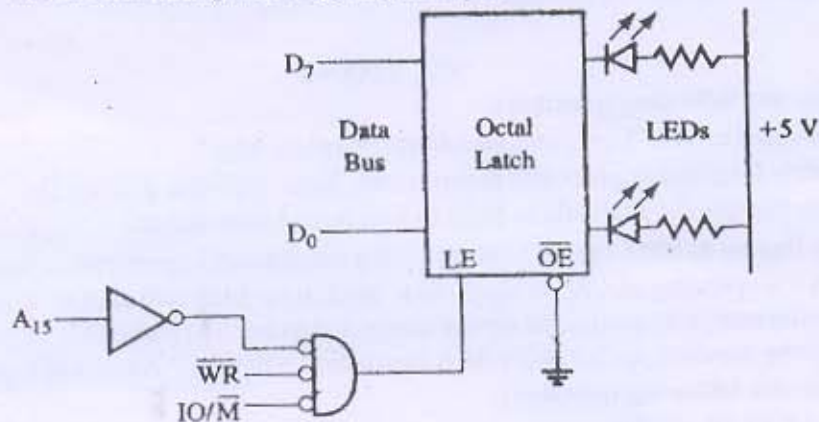
OR

[C] If 8085 is not capable of reading the data from fast peripheral device, what technique is used ? Explain the support available from 8085. [7]

Q-6 Answer the following questions.(Attempt any Two) [12]

[A] Explain Various addressing modes giving example.

[B]



Identify the port address of the output port in the above figure. Write a program to glow each LED one by one continuously.

[C] Write down the differences between memory-mapped I/O and peripheral I/O.

[D] Compare: 8086, 80286, 80386.